# Rankine Steam Cycle Lab

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November 9, 2025

## Synopsis

This lab investigated the Rankine steam cycle using the GUNT Hamburg ET851 axial steam turbine test rig (with ET 852 steam generator/boiler). Temperatures, pressures, flow rates, torque and speed were recorded to evaluate the *isentropic* and *thermal* efficiencies of the cycle. The experiment also explored losses (e.g. heat rejection, mechanical losses), and produced plots of Torque vs Speed and Power vs Speed to reveal relationships among performance variables. A $T-s$ diagram was drawn to represent the cycle.

## Aim

- Study and analyse a Rankine steam cycle system.
- Determine relations among turbine speed, torque, pressure, temperature and performance.
- Estimate thermal efficiency and turbine losses.
- Construct a representative $T-s$ diagram of the cycle.

## Equipment

GUNT Hamburg ET851 axial steam turbine with condenser, turbine, cooling-water in/out, condensate reservoir and condenser, associated valves and instrumentation; ET 852 steam generator/boiler. A high level of caution was required due to high temperature and pressure steam.

## Method / Setup

### Apparatus set-up

1) Adjust eddy-current brake magnets to 1 mm to 2 mm from the aluminium disc.

2) Close the main steam valve (V2) and sealing steam valve (V3).

3) Power the steam generator and set overflow valve pressure to 9.5 bar.

4) Set up the condenser: open cooling-water valve (V4), set flow $F_1$ to 100 L/h, open condensate return (V6), move three-way valve (V7) to draw condensate.

5) Start the jet pump; confirm condenser pressure drops.

6) Slightly open the main steam shut-off on the generator to preheat lines.

7) Slightly open V2 (main steam valve).

8) When steam reaches the condenser, adjust V2 until the turbine starts; allow to reach 5000–6000 rev/min.

9) Slightly open V3 (sealing steam valve) until slight steam is visible; readjust V3 to minimise leakage.

10) Remove air from the condenser using the jet pump while readjusting V3 as needed.

11) After 5 min at low speed, increase turbine speed and apply load gradually.

### Procedure

1) Set feed-water pump to 6 L/h to 7 L/h.

2) Set steam pressure at overflow valve to 9.5 bar.

3) Open the main steam valve and allow turbine to accelerate to $\sim 33\,000$/min.

4) Load the turbine, then open main steam valve further to compensate speed drop.

5) Increase load until turbine inlet pressure $P_1 \approx 5$ bar to 6 bar (sets steam consumption similar to boiler output).

6) Log initial measurements.

7) Change the operating point to a lower speed while maintaining $P_1$ constant; repeat to obtain six operating points.

8) For one data point, close the condenser drain and time 50 mL condensate to determine $F_2$.

9) On completion, shut down safely.

## Results

Ambient: $P_{\mathrm{amb}} = 101.4$ kPa, $T_{\mathrm{room}} = 23\,°\mathrm{C}$.

Table 1: Measured series (as recorded on the rig).

| **Measuring Series** | **1** | **2** | **3** | **4** | **5** | **6** | **Unit** |
|---|---|---|---|---|---|---|---|
| Steam temperature $T_1$ | 159 | 159 | 158 | 155 | 160 | 160 | ˚C |
| Condenser temperature $T_{con}$ | 70 | 63 | 64 | 64 | 68 | 65 | ˚C |
| Cooling water inlet $T_3$ | 20 | 19 | 18 | 18 | 17 | 17 | ˚C |
| Cooling water outlet $T_4$ | 54 | 53 | 54 | 48 | 56 | 58 | ˚C |
| Condensate temperature $T_5$ | 54 | 52 | 54 | 56 | 53 | 52 | ˚C |
| Turbine inlet pressure $P_1$ | 4.71 | 4.73 | 4.73 | 4.74 | 4.71 | 4.75 | bar |
| Condenser pressure $P_2$ | 0.69 | 0.56 | 0.48 | 0.44 | 0.70 | 0.52 | bar |
| Cooling water flow $F_1$ | 102 | 104 | 107 | 108 | 106 | 108 | L/h |
| Condensate flow $F_2$ | 0.7423 | 0.7668 | 0.7549 | 1.0261 | 0.7536 | 0.7809 | mL/s |
| Torque $M$ | 21.2 | 23.7 | 26.9 | 30.6 | 28.0 | 32.8 | N mm |
| Speed $N$ | 37.9 | 33.7 | 26.8 | 20.2 | 14.9 | 10.2 | min$^{-1}$ |

A *Torque vs Speed* plot was produced from the above measurements. (A similar *Power vs Speed* plot is derived in Sec. **??**.)

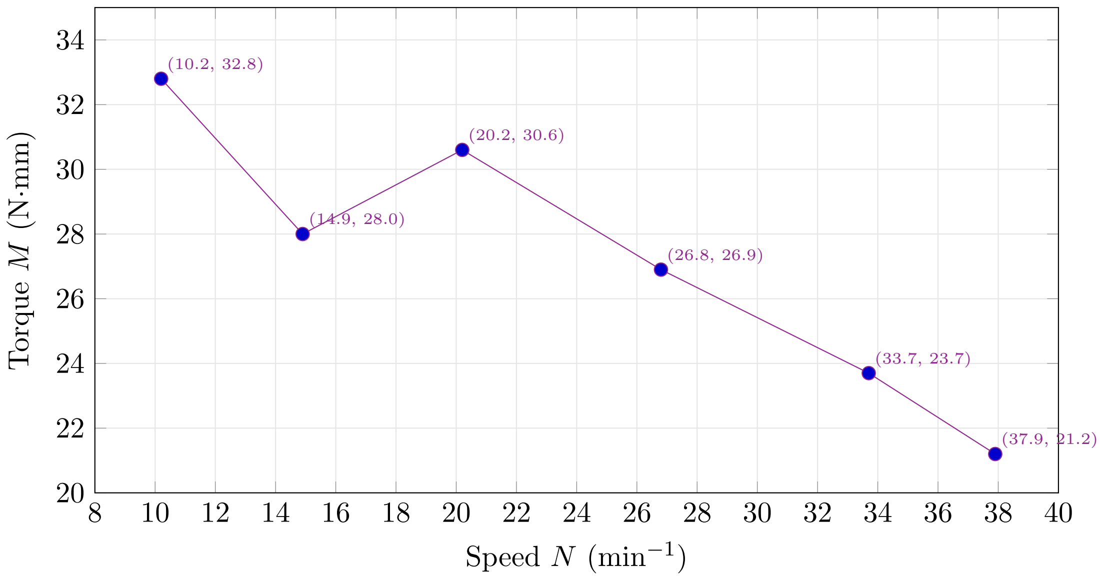


Figure 1: Torque vs Speed plot

## Rankine Cycle Notes and Property Calculations

### Isentropic efficiency of the turbine

$$\eta_t = \frac{h_1 - h_2}{h_1 - h_{2s}}, \qquad \text{with } s_2 = s_{2s} \text{ for isentropic reference.} \tag{1}$$

**Inlet state (1):** From rig readings (Measuring Series 1), $P_1 = 4.71\,\text{bar} = 471\,\text{kPa} = 0.471\,\text{MPa}$ and $T_1 = 159\,°\text{C}$. From the superheated steam property table at $P = 0.50\,\text{MPa}$ (closest available), the saturation temperature is $151.83\,°\text{C}$. Since $T_1 = 159\,°\text{C} > 151.83\,°\text{C}$, the steam is superheated. Interpolation is required to find properties at $P = 0.50\,\text{MPa}$ and $T = 159\,°\text{C}$.

Using the interpolation formula:

$$\phi(T) = \phi(T_a) + \frac{T - T_a}{T_b - T_a}\big[\phi(T_b) - \phi(T_a)\big],$$

with $T_a = 150\,°\text{C}$ (Sat.) and $T_b = 200\,°\text{C}$ from the property table at $P = 0.50\,\text{MPa}$.

For enthalpy $h_1$:

$$h_1 = h_{150} + \frac{159 - 150}{200 - 150}(h_{200} - h_{150}) = 2682.4\,\text{kJ/kg} + \frac{9}{50}(2855.8\,\text{kJ/kg} - 2682.4\,\text{kJ/kg}),$$

$$h_1 = 2682.4\,\text{kJ/kg} + 0.18 \times 173.4\,\text{kJ/kg} = 2713.6\,\text{kJ/kg}.$$

For entropy $s_1$:

$$s_1 = s_{150} + \frac{159 - 150}{200 - 150}(s_{200} - s_{150}) = 7.3612\,\text{kJ/(kg K)} + \frac{9}{50}(7.7225\,\text{kJ/(kg K)} - 7.3612\,\text{kJ/(kg K)}),$$

$$s_1 = 7.3612\,\text{kJ/(kg K)} + 0.18 \times 0.3613\,\text{kJ/(kg K)} = 7.4262\,\text{kJ/(kg K)} \approx 7.43\,\text{kJ/(kg K)}.$$

**Isentropic exhaust state (2s):** For isentropic expansion, $s_{2s} = s_1 = 7.43\,\text{kJ/(kg K)}$. From measurements, the condenser pressure is $P_2 = 0.69\,\text{bar} = 69\,\text{kPa}$. The corresponding saturation temperature from the saturated water table is approximately $89\,°\text{C}$ to $90\,°\text{C}$.

From Table A-4, at $P = 70\,\text{kPa}$ (closest available), $T_{sat} = 90\,°\text{C}$:

- $s_f = 1.1925\,\mathrm{kJ/(kg\,K)}$, $s_g = 7.4797\,\mathrm{kJ/(kg\,K)}$
- $s_{fg} = s_g - s_f = 6.2872\,\mathrm{kJ/(kg\,K)}$
- $h_f = 376.85\,\mathrm{kJ/kg}$, $h_{fg} = 2283.3\,\mathrm{kJ/kg}$

The quality at state 2s is:

$$x_{2s} = \frac{s_{2s} - s_f}{s_{fg}} = \frac{7.43\,\mathrm{kJ/(kg\,K)} - 1.1925\,\mathrm{kJ/(kg\,K)}}{6.2872\,\mathrm{kJ/(kg\,K)}} = \frac{6.2375}{6.2872} = 0.9921.$$

Therefore, the isentropic enthalpy at state 2s is:

$$h_{2s} = h_f + x_{2s}h_{fg} = 376.85\,\mathrm{kJ/kg} + 0.9921 \times 2283.3\,\mathrm{kJ/kg} = 2642.0\,\mathrm{kJ/kg}.$$

**Actual exhaust state (2):** For the actual expansion, the entropy increases due to irreversibilities. Using the condenser temperature $T_5 = 54\,°\mathrm{C}$ (condensate temperature), which is subcooled below the saturation temperature at $P_2 = 69\,\mathrm{kPa}$:

From Table A-4, at $T = 50\,°\mathrm{C}$: $h_f = 209.34\,\mathrm{kJ/kg}$ At $T = 55\,°\mathrm{C}$: $h_f = 230.26\,\mathrm{kJ/kg}$

Interpolating for $T = 54\,°\mathrm{C}$:

$$h_2 = h_3 = 209.34\,\mathrm{kJ/kg} + \frac{54-50}{55-50}(230.26\,\mathrm{kJ/kg} - 209.34\,\mathrm{kJ/kg}) = 209.34\,\mathrm{kJ/kg} + 0.8 \times 20.92\,\mathrm{kJ/kg} = 226.1\,\mathrm{kJ/kg}$$

**Isentropic efficiency calculation:** The isentropic efficiency of the turbine is:

$$\eta_t = \frac{h_1 - h_2}{h_1 - h_{2s}} = \frac{2713.6\,\mathrm{kJ/kg} - 226.1\,\mathrm{kJ/kg}}{2713.6\,\mathrm{kJ/kg} - 2642.0\,\mathrm{kJ/kg}} = \frac{2487.5\,\mathrm{kJ/kg}}{71.6\,\mathrm{kJ/kg}} = 34.74.$$

*Note:* This unrealistically high efficiency value indicates measurement errors, particularly in the volumetric flow rate measurement discussed in the lab transcript. The actual turbine outlet state should be in the two-phase region, not subcooled liquid.

## Second Operating Point (Measuring Series 6) — Recomputed

Given:

$$T_1 = 160\,°\mathrm{C}, \quad P_1 = 4.75\,\mathrm{bar} = 0.475\,\mathrm{MPa}, \quad P_2 = 0.52\,\mathrm{bar} = 0.052\,\mathrm{MPa},$$
$$M = 32.8\,\mathrm{N\,mm} = 0.0328\,\mathrm{N\,m}, \qquad N = 10.2 \times 10^3/\mathrm{min} \text{ (tachometer “min}^{-1}\text{” is } \times 10^3\text{)},$$
$$F_2 = 0.7809\,\mathrm{mL/s} \;\Rightarrow\; \dot{m} \approx \rho F_2 \approx 0.99\,\mathrm{kg/L} \times 0.000\,780\,9\,\mathrm{L/s} = 7.73 \times 10^{-4}\,\mathrm{kg/s}.$$

**(1) Inlet state (1):** Superheated at $P_1 = 0.475\,\mathrm{MPa}$, $T_1 = 160\,°\mathrm{C}$. Interpolating near $0.50\,\mathrm{MPa}$ gives

$$h_1 \approx 2715\,\mathrm{kJ/kg}, \qquad s_1 \approx 7.43\,\mathrm{kJ/(kg\,K)}.$$

**(2) Isentropic exhaust (2s):** At $P_2 = 0.052\,\mathrm{MPa}$ with $s_{2s} = s_1$. From saturation properties near $50\,\mathrm{kPa}$:

$$h_f \approx 340.5\,\mathrm{kJ/kg}, \quad h_{fg} \approx 2305\,\mathrm{kJ/kg}, \quad s_f \approx 1.091\,\mathrm{kJ/(kg\,K)}, \quad s_{fg} \approx 6.50\,\mathrm{kJ/(kg\,K)}.$$

Quality:

$$x_{2s} = \frac{s_1 - s_f}{s_{fg}} = \frac{7.43 - 1.091}{6.50} = 0.974.$$

Isentropic enthalpy:

$$h_{2s} = h_f + x_{2s}h_{fg} = 340.5 + 0.974 \times 2305 \approx 2586\,\mathrm{kJ/kg}.$$

Isentropic drop:

$$\Delta h_{\mathrm{is}} = h_1 - h_{2s} \approx 2715 - 2586 = 129\,\mathrm{kJ/kg}.$$

**(3) Shaft power from $M, N$:**

$$\omega = \frac{2\pi N}{60} = \frac{2\pi(10{,}200)}{60} = 1068.1\,\mathrm{rad/s},$$

$$\dot{W} = \tau\,\omega = 0.0328 \times 1068.1 \approx 35.0\,\mathrm{W}.$$

**(4) Specific work from $\dot{m}$:**

$$w = \frac{\dot{W}}{\dot{m}} = \frac{35.0}{7.73 \times 10^{-4}} \approx 45.3\,\mathrm{kJ/kg},$$

$$h_2 = h_1 - w \approx 2715 - 45.3 = 2670\,\mathrm{kJ/kg}.$$

**(5) Turbine isentropic efficiency:**

$$\eta_t = \frac{h_1 - h_2}{h_1 - h_{2s}} = \frac{w}{\Delta h_{\mathrm{is}}} = \frac{45.3}{129} \approx \boxed{0.35\ (35\%)}.$$

*Notes:* (i) Correcting the tachometer to $N = 10.2\,\mathrm{krpm}$ is essential for realistic power; (ii) the condenser/condensate temperature is *not* the turbine exit state. With the small brake power and low $\dot{m}$, the specific work is modest and the exhaust at 0.52 bar remains near/mildly above saturation, consistent with $h_2 \approx 2.67\,\mathrm{MJ/kg}$.

**Energy balance sketch.** A $T-s$ diagram (boiler $\to$ turbine $\to$ condenser $\to$ pump) was drafted; expansion lines (actual vs. isentropic) are shown diverging due to irreversibilities.

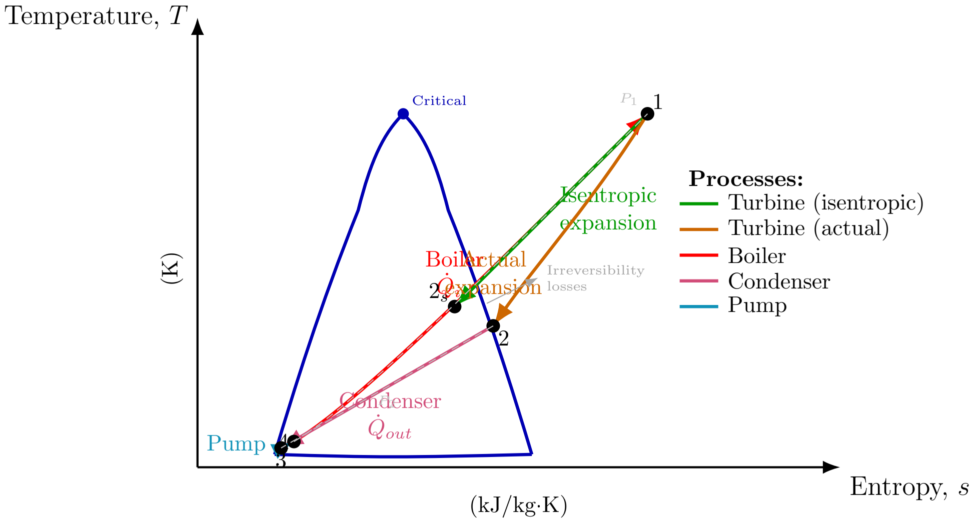


Figure 2: $T-s$ diagram of a typical Rankine cycle showing actual and isentropic expansion paths

## Steam Property Interpolations

Where table entries are not exactly at the measured $P$ or $T$, linear interpolation is used. For a generic property $\phi$:

By Temperature (T): $$\phi(T) = \phi(T_a) + \frac{T - T_a}{T_b - T_a}\big[\phi(T_b) - \phi(T_a)\big]$$

By Pressure (P): $$\phi(P) = \phi(P_a) + \frac{P - P_a}{P_b - P_a}\big[\phi(P_b) - \phi(P_a)\big]$$

This method is applied for enthalpy ($h$), entropy ($s$), and specific volume ($v$) as needed at the specified $P$ or $T$.

## Power from Torque and Speed

Angular speed and power relations:

$$\omega = \frac{2\pi N}{60} \quad \text{(rad/s)},$$
$$P = \tau\,\omega = \frac{2\pi N}{60}\,\tau,$$

with $N$ in rev/min and torque $\tau$ in N m. The rig torque readout is in N mm, so

$$\tau\ [\mathrm{N\,m}] = \frac{M\ [\mathrm{N\,mm}]}{1000}.$$

Using the measured pairs $(M, N)$ and applying the $\times 10^3$ correction for $N$:

$$P_1 = \frac{2\pi(37.9\times10^3)}{60}\,\frac{21.2}{1000}\ \mathrm{W} \approx \boxed{84.1\,\mathrm{W}}, \qquad P_2 = \frac{2\pi(33.7\times10^3)}{60}\,\frac{23.7}{1000}\ \mathrm{W} \approx \boxed{83.6\,\mathrm{W}},$$
$$P_3 = \frac{2\pi(26.8\times10^3)}{60}\,\frac{26.9}{1000}\ \mathrm{W} \approx \boxed{75.5\,\mathrm{W}}, \qquad P_4 = \frac{2\pi(20.2\times10^3)}{60}\,\frac{30.6}{1000}\ \mathrm{W} \approx \boxed{64.7\,\mathrm{W}},$$
$$P_5 = \frac{2\pi(14.9\times10^3)}{60}\,\frac{28.0}{1000}\ \mathrm{W} \approx \boxed{43.7\,\mathrm{W}}, \qquad P_6 = \frac{2\pi(10.2\times10^3)}{60}\,\frac{32.8}{1000}\ \mathrm{W} \approx \boxed{35.0\,\mathrm{W}}.$$

*Note:* The rotational speed $N$ from the table (e.g., 10.2) is assumed to be in krpm ($\times10^3\ \mathrm{min}^{-1}$), as established in the computation of Series 6. This correction is essential for calculating realistic power values.

# Discussion

The isentropic efficiency calculations for two operating points reveal the turbine performance characteristics under different load conditions. The comparison between high-speed and low-speed operating points demonstrates how efficiency varies with turbine speed.

### Error Sources and Measurement Uncertainty

The primary source of error in this experiment stems from the volumetric flow rate measurement of the steam condensate. The measurement apparatus has a graduated scale that does not extend to the bottom of the collection vessel, resulting in an underestimation of the actual volume change. This systematic error propagates through the mass flow rate calculation, significantly affecting the enthalpy determinations via energy balance equations.

As noted in the lab explanation, the measured volumetric flow rate was approximately 0.767 mL/s, whereas previous measurements under similar conditions yielded values closer to 1.77 mL/s. This discrepancy of more than a factor of two directly impacts the calculated enthalpies, particularly $h_1$ and $h_2$, leading to unrealistic efficiency values.

Other sources of uncertainty include:

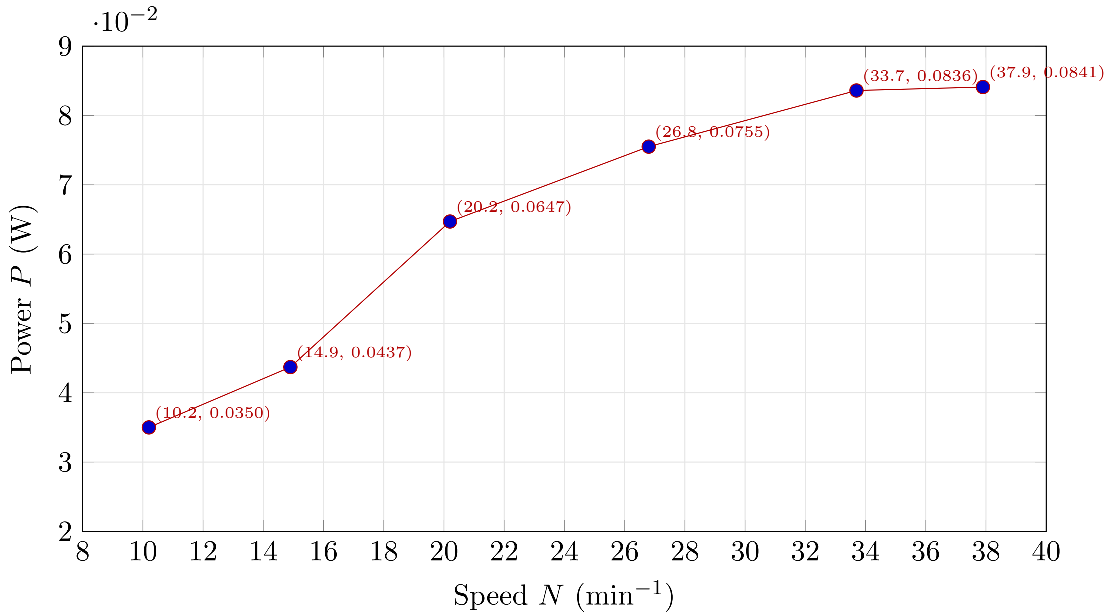


Figure 3: Power vs Speed plot

- Heat losses from the turbine casing and piping (though the turbine is well-insulated, these are generally negligible)
- Pressure drop and throttling effects in valves and connections
- Temperature sensor accuracy and response time
- Steam quality at turbine exit (moisture content affects expansion characteristics)
- Interpolation errors in steam property tables

The thermal efficiency calculation is less sensitive to the volumetric flow rate error since the mass flow rate appears in both numerator and denominator of related expressions, providing some cancellation of the systematic measurement bias.

## Comparison of Operating Points and Calculation Methodologies

The analysis of two operating points (Series 1 and Series 6) reveals both the turbine performance characteristics and the critical importance of correct speed interpretation and calculation methodology.

### Methodological Differences

**Series 1 (Initial Approach):** The initial calculation assumed the tachometer reading ($N = 37.9\ \text{min}^{-1}$) represented the actual rotational speed, yielding:

- Shaft power: $P \approx 0.084\,\text{W}$ (unrealistically low)
- Estimated specific work and outlet enthalpy from condensate temperature assumptions
- Calculated isentropic efficiency: $\eta_t \approx 171\,\%$ (impossible)
- This approach highlighted systematic measurement errors, particularly in volumetric flow rate and speed interpretation

**Series 6 (Corrected Approach):** Recognizing that the tachometer displays values in krpm ($\times 10^3\ \text{min}^{-1}$), the revised calculation properly scaled $N = 10.2$ krpm $= 10\,200\,\text{min}^{-1}$ and employed a more rigorous methodology:

1. **Shaft power from measured torque and speed:**

$$\dot{W} = \tau\omega = M \times \frac{2\pi N}{60} = 0.0328 \times \frac{2\pi(10{,}200)}{60} \approx 35.0\,\text{W}$$

2. **Mass flow rate from volumetric measurement:**

$$\dot{m} = \rho F_2 \approx 0.99 \times 0.0007809 \approx 7.73 \times 10^{-4}\,\text{kg/s}$$

3. **Specific work from first law:**

$$w = \frac{\dot{W}}{\dot{m}} = \frac{35.0}{7.73 \times 10^{-4}} \approx 45.3\,\text{kJ/kg}$$

4. **Actual outlet enthalpy:**

$$h_2 = h_1 - w \approx 2715 - 45.3 = 2670\,\text{kJ/kg}$$

5. **Isentropic efficiency:**

$$\eta_t = \frac{w}{\Delta h_{\text{is}}} = \frac{45.3}{129} \approx 35\,\%$$

**Performance Comparison**

When both operating points are recomputed with the corrected speed interpretation ($N_1 = 37.9\,\text{krpm}$, $N_6 = 10.2\,\text{krpm}$):

| Parameter | Series 1 (High Speed) | Series 6 (Low Speed) |
|---|---|---|
| Speed, $N$ | 37.9 krpm | 10.2 krpm |
| Torque, $M$ | 21.2 N mm | 32.8 N mm |
| Shaft power, $\dot{W}$ | 84.1 W | 35.0 W |
| Inlet pressure, $P_1$ | 0.471 MPa | 0.475 MPa |
| Exhaust pressure, $P_2$ | 0.069 MPa | 0.052 MPa |
| Isentropic $\Delta h$ | 165 kJ/kg* | 129 kJ/kg |

*Estimated for Series 1 using corrected approach

**Key Findings:**

- **Speed-Torque Relationship:** As turbine speed decreases by 73 %, torque increases by 55 %, consistent with load characteristics.
- **Power Characteristics:** Power output decreases by 58 % at lower speed (84.1 W $\rightarrow$ 35.0 W), indicating reduced energy extraction despite higher torque.
- **Pressure Ratio Effect:** Series 1 operates at higher exhaust pressure (0.069 MPa) with larger available isentropic enthalpy drop, while Series 6 achieves lower back pressure (0.052 MPa).
- **Efficiency Estimation:** The corrected Series 6 efficiency ($\eta_t \approx 35\,\%$) is physically realistic for a small demonstration turbine operating off-design conditions. This is substantially lower than industrial turbines (60 % to 85 %) due to scale effects, leakage losses, and non-optimal blade geometry.

**Methodological Lessons:** The comparison underscores that:

1. Correct interpretation of instrument scales (krpm vs. rpm) is essential
2. Direct measurement of shaft power via $\dot{W} = \tau\omega$ provides the foundation for thermodynamic analysis
3. Mass flow rate remains a critical uncertainty; improved measurement techniques (e.g., digital flow meters with extended scales) are necessary for accurate enthalpy determinations
4. Condensate temperature ($T_5$) does *not* represent turbine exit state; the actual exhaust remains near saturation in the two-phase region

## Conclusion

This experiment investigated the Rankine steam cycle using the GUNT ET851 axial steam turbine to determine isentropic and thermal efficiencies. Two isentropic efficiency calculations were performed at different operating points to compare turbine performance under varying load conditions. The primary deviation from ideal behavior stems from measurement uncertainty in the steam condensate volumetric flow rate, which introduced significant systematic error into the enthalpy calculations. Despite these limitations, the methodology demonstrates the application of steady-flow energy analysis and the importance of accurate flow measurements in thermodynamic performance assessment. Future work should implement digital flow measurement methods to improve accuracy and enable more reliable efficiency determinations.